Memory Stability Log

Iteration	Action	Network	Buffer	Sent	Dropped	Duplicates	Free heap	Notes
0	Baseline	CONNECTED	0	0	0	0	350868	baseline after RESET STATS
1	STRESS 100	CONNECTED	0	100	0	0	342604	−8,264 bytes (≈82.6 B/msg); slightly larger, likely one-time init
2	STRESS 100	CONNECTED	0	200	0	0	334604	−8,000 bytes (80 B/msg)
3	STRESS 100	CONNECTED	0	300	0	0	326612	−7,992 bytes (≈79.9 B/msg)
4	STRESS 100	CONNECTED	0	400	0	0	318600	−8,012 bytes (≈80.1 B/msg)
5	STRESS 100	CONNECTED	0	500	0	0	310600	−8,000 bytes (80 B/msg)
6	STRESS 100	CONNECTED	0	600	0	0	302600	−8,000 bytes (80 B/msg)
7	STRESS 100	CONNECTED	0	700	0	0	294600	−8,000 bytes (80 B/msg)
8	STRESS 100	CONNECTED	0	800	0	0	286600	−8,000 bytes (80 B/msg)
9	STRESS 100	CONNECTED	0	900	0	0	278600	−8,000 bytes (80 B/msg)

10	STRESS 100	CONNECTED	0	1000	0	0	270600	Total −80,268 bytes, ≈80 B/message, heap does not recover.

Heap decreased by ≈8,000 bytes (≈80 B per message) after every STRESS 100 iteration and never recovered: 350,868 → 270,600 (−80,268 bytes, −22.9%) over 1,000 messages, with buffer=0, dropped=0, duplicates=0. This is a systematic, irreversible decline, which violates R7 (memory leak).

STRESS during NET DOWN

Iteration	Action	Network	Buffer	Sent	Dropped	Duplicates	Free heap	Notes
0	Baseline	CONNECTED	0	0	0	0	350868	baseline after RESET STATS
1	NET DOWN, STRESS 100, NET UP	CONNECTED	0	200	0	100	342612	−8,256 B (≈82.6 B/msg); 100 generated, 200 sent; reconnects: 84
2	NET DOWN, STRESS 100, NET UP	CONNECTED	0	400	0	200	334612	−8,000 B (80 B/msg); +200 sent, +100 dup; reconnects: 147
3	NET DOWN, STRESS 100, NET UP	CONNECTED	0	600	0	300	326612	−8,000 B (80 B/msg); +200 sent, +100 dup; reconnects: 215
4	NET DOWN, STRESS 100, NET UP	CONNECTED	0	800	0	400	318600	−8,012 B (≈80.1 B/msg); +200 sent, +100 dup;

								reconnects: 260
5	NET DOWN, STRESS 100, NET UP	CONNECTED	0	1000	0	500	310600	−8,000 B (80 B/msg); total −40,268 B (≈80.5 B/msg), heap does not recover; reconnects: 357

STRESS 100 under NET DOWN (5 runs): each run generated 100 messages, but `sent` increased by 200 and `duplicates` by 100, so every message was published twice after NET UP (R6 FAIL). `dropped` stayed 0 although 100 messages exceeded capacity 8 (R4 FAIL). Free heap dropped by ≈8,000 B per run (≈80 B/msg), same as online, and never recovered (R7 FAIL). Reconnect attempts per run varied (45–97) depending on how long NET DOWN lasted.